

physical interpretations and our notation must clearly distinguish between them. The absence of frequency information in the phasors results from the fact that in the sinusoidal steady state, all currents and voltages are sinusoids with the same frequency. Carrying frequency information in the phasor would be redundant, since it is the same for all phasors in any given steady-state circuit problem.

In summary, given a sinusoidal signal $v(t) = V_A \cos(\omega t + \phi)$, the corresponding phasor representation is $\mathbf{V} = V_A e^{j\phi} = V_A \angle \phi$. Conversely, given the phasor $\mathbf{V} = V_A e^{j\phi}$, the corresponding sinusoid is found by multiplying the phasor by $e^{j\omega t}$ and reversing the steps in Eq. (8-4) as follows:

$$\begin{aligned} v(t) &= \operatorname{Re}\{\mathbf{V}e^{j\omega t}\} = \operatorname{Re}\{(V_A e^{j\phi})e^{j\omega t}\} \\ &= V_A \operatorname{Re}\{e^{j(\omega t + \phi)}\} = V_A \operatorname{Re}\{\cos(\omega t + \phi) + j \sin(\omega t + \phi)\} \\ &= V_A \cos(\omega t + \phi) \end{aligned} \quad (8-6)$$

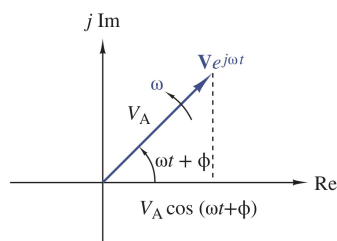


FIGURE 8-2 Complex exponential $\mathbf{V}e^{j\omega t}$.

The frequency ω in the complex exponential $\mathbf{V}e^{j\omega t}$ in Eq. (8-6) must be expressed or implied in a problem statement, since by definition it is not contained in the phasor. Figure 8-2 shows a geometric interpretation of the complex exponential $\mathbf{V}e^{j\omega t}$ as a vector in the complex plane of length V_A , which rotates counterclockwise with a constant angular velocity of ω , the radian frequency of the sinusoid. The real part operation projects the rotating vector onto the horizontal (real) axis and thereby generates $v(t) = V_A \cos(\omega t + \phi)$. The complex exponential is sometimes called a **rotating phasor**, and the phasor $\mathbf{V}$ is viewed as a snapshot of the situation at $t = 0$.

Exercise 8-1

Convert the following sinusoids to phasors in polar and rectangular form:

- (a) $v(t) = 20 \cos(150t - 60^\circ)$ V
- (b) $v(t) = 10 \cos(1000t + 180^\circ)$ V
- (c) $i(t) = -4 \cos 3t + 3 \cos(3t - 90^\circ)$ A

Answers:

- (a) $\mathbf{V} = 20 \angle -60^\circ = 10 - j17.3$ V
- (b) $\mathbf{V} = 10 \angle 180^\circ = -10 + j0$ V
- (c) $\mathbf{I} = 5 \angle -143^\circ = -4 - j3$ A

Exercise 8-2

Convert the following phasors to sinusoids:

- (a) $\mathbf{V} = 169 \angle -45^\circ$ V at $f = 60$ Hz
- (b) $\mathbf{V} = 10 \angle 90^\circ + 66 - j10$ V at $\omega = 10$ krad/s
- (c) $\mathbf{I} = 15 + j5 + 10 \angle 180^\circ$ mA at $\omega = 1000$ rad/s

Answers:

- (a) $v(t) = 169 \cos(377t - 45^\circ)$ V
- (b) $v(t) = 66 \cos 10^4 t$ V
- (c) $i(t) = 7.07 \cos(1000t + 45^\circ)$ mA

PROPERTIES OF PHASORS

Two important properties of phasors play key roles in circuit analysis. First, the **additive property** states that the phasor representing a sum of sinusoids of the same frequency is obtained by adding the phasor representations of the component sinusoids. To establish this property we write the expression

$$\begin{aligned}
 v(t) &= v_1(t) + v_2(t) + \cdots + v_N(t) \\
 &= \operatorname{Re}\{\mathbf{V}_1 e^{j\omega t}\} + \operatorname{Re}\{\mathbf{V}_2 e^{j\omega t}\} + \cdots + \operatorname{Re}\{\mathbf{V}_N e^{j\omega t}\}
 \end{aligned}
 \quad (8-7)$$

where $v_1(t), v_2(t), \dots, v_N(t)$ are sinusoids of the same frequency whose phasor representations are $\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_N$. The real part operation is additive, so the sum of real parts equals the real part of the sum. Consequently, Eq. (8-7) can be written in the form

$$\begin{aligned}
 v(t) &= \operatorname{Re}\{\mathbf{V}_1 e^{j\omega t} + \mathbf{V}_2 e^{j\omega t} + \cdots + \mathbf{V}_N e^{j\omega t}\} \\
 &= \operatorname{Re}\{(\mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_N) e^{j\omega t}\}
 \end{aligned}
 \quad (8-8)$$

Comparing the last line in Eq. (8-8) with the definition of a phasor, we conclude that the phasor $\mathbf{V}$ representing $v(t)$ is

$$\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2 + \cdots + \mathbf{V}_N \quad (8-9)$$

The result in Eq. (8-9) applies only if the component sinusoids all have the same frequency so that $e^{j\omega t}$ can be factored out as shown in the last line in Eq. (8-8).

In Chapter 5 we found that the time derivative of a sinusoid is another sinusoid with the same frequency. Since they have the same frequency, the signal and its derivative can be represented by phasors. The **derivative property** of phasors allows us easily to relate the phasor representing a sinusoid to the phasor representing its derivative.

Equation (8-6) relates a sinusoid function and its phasor representation as

$$v(t) = \operatorname{Re}\{\mathbf{V} e^{j\omega t}\}$$

Differentiating this equation with respect to time t yields

$$\begin{aligned}
 \frac{dv(t)}{dt} &= \frac{d}{dt} \operatorname{Re}\{\mathbf{V} e^{j\omega t}\} = \operatorname{Re}\left\{\mathbf{V} \frac{d}{dt} e^{j\omega t}\right\} \\
 &= \operatorname{Re}\{(j\omega \mathbf{V}) e^{j\omega t}\}
 \end{aligned}
 \quad (8-10)$$

From the definition of a phasor, we see that the quantity $(j\omega \mathbf{V})$ on the right side of this equation is the phasor representation of the time derivative of the sinusoidal waveform. This phasor can be written in the form

$$\begin{aligned}
 j\omega \mathbf{V} &= (\omega e^{j90^\circ}) (V_A e^{j\theta}) \\
 &= \omega V_A e^{j(\theta+90^\circ)}
 \end{aligned}
 \quad (8-11)$$

which points out that differentiating a sinusoid changes its amplitude by a multiplicative factor ω and shifts the phase angle by 90° .

In summary, the additive property states that adding phasors is equivalent to adding sinusoids of the same frequency. The derivative property states that multiplying a phasor by $j\omega$ is equivalent to differentiating the corresponding sinusoid. The following examples show applications of these two properties of phasors.



EXAMPLE 8-1

(a) Construct the phasors for the following signals:

$$\begin{aligned}
 v_1(t) &= 10 \cos(1000t - 45^\circ) \text{ V} \\
 v_2(t) &= 5 \cos(1000t + 30^\circ) \text{ V}
 \end{aligned}$$

(b) Use the additive property of phasors and the phasors found in (a) to find $v(t) = v_1(t) + v_2(t)$.